

Swap Pricing Formulae

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n	number of fixed coupons
m	number of floating coupons
p	par rate
l_i	i th LIBOR forecast rate
s	floating spread over libor
N	swap notional
$P(t_0, t_i)$	discount factor for the i th coupon
ϕ	IRS Indicator Function +1 for a receiver swap and -1 for a payer swap
ϕ	Tenor Basis Indicator Function +1 when receiving 6m Libor and -1 when paying
τ_i	i th accrual period or year fraction
t_0	time to the valuation date in years, usually zero
t_i	time to the i th coupon payment date in years

Annuity

$$\text{Annuity}^{\text{Fixed}} = \sum_{i=1}^n \tau_i P(t_0, t_i) \quad (1)$$

Swap PV

$$\begin{aligned} PV(\text{Swap}) &= \phi \left(PV(\text{Fixed}) - PV(\text{Float}) \right) \\ &= \phi N \left(r^{\text{Fixed}} \sum_{i=1}^n \tau_i P(t_0, t_i) - \sum_{j=1}^m (l_j + s) \tau_j P(t_0, t_j) \right) \\ &= \phi N \left(r^{\text{Fixed}} \text{Annuity}^{\text{Fixed}} - \sum_{j=1}^m (l_j + s) \tau_j P(t_0, t_j) \right) \end{aligned} \quad (2)$$

Tenor Basis Swap PV

$$\begin{aligned} PV(\text{Tenor Basis } 3 \times 6) &= \phi \left(PV(\text{FloatLeg}^{6m\text{Libor}}) - PV(\text{FloatLeg}^{3m\text{Libor}}) \right) \\ &= \phi \left(\sum_{i=1}^n l_i^{6m} \tau_i^{6m} P(t_0, t_i) - \sum_{j=1}^m (l_j^{3m} + s) \tau_j^{3m} P(t_0, t_j) \right) \end{aligned} \quad (3)$$

Par Rate

$$ParRate = \frac{\sum_{j=1}^m l_j \tau_j P(t_0, t_j)}{\sum_{i=1}^n \tau_i P(t_0, t_i)} = \frac{PV(Float)}{Annuity^{Fixed}} \quad (4)$$

Basis Spread

$$\begin{aligned} s &= \frac{\left(\sum_{i=1}^n l_i^{6m} \tau_i^{6m} P(t_0, t_i) - \sum_{j=1}^m l_j^{3m} \tau_j^{3m} P(t_0, t_j) \right)}{\sum_{j=1}^m (l_j^{3m} + s) \tau_j^{3m} P(t_0, t_j)} \\ &= \frac{\left(\sum_{i=1}^n l_i^{6m} \tau_i^{6m} P(t_0, t_i) - \sum_{j=1}^m l_j^{3m} \tau_j^{3m} P(t_0, t_j) \right)}{Annuity^{3m}} \end{aligned} \quad (5)$$

The End